

Trial-Guard: Clinical Trial Risk & Protocol Deviation Monitor Protocol Deviation & Site Risk Monitor

Real-time protocol compliance and site-level risk monitoring

Hackathon Project | IBM Industry Track

Team

NEXORA

Study

ONCO-4471 · Phase
III

Scope

250 synthetic sites

The Problem

Protocol deviations can affect patient safety, data integrity, and study quality.

What is happening?

Patient visit records must be checked against protocol rules such as visit windows, dosing, medication restrictions, assessments, eligibility and consent.

Who needs it?

Clinical operations teams, site coordinators and study oversight teams need a fast way to identify problematic sites.

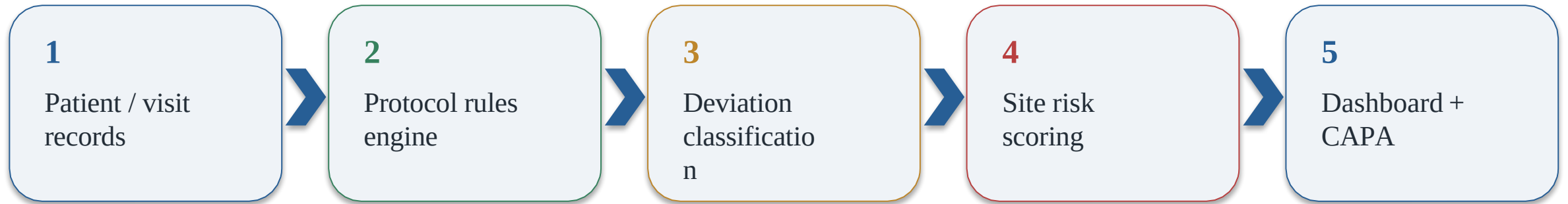
Why it matters

Manual review can make it harder to spot worsening trends and prioritize sites that need corrective action.

- Need: consistent severity classification
- Need: site-level risk prioritization
- Need: actionable CAPA recommendations

Solution

A rule-based monitoring dashboard converts visit records into prioritized site risk.



- Filters by site, region and risk tier
- Opens site-level deviation details
- Generates CAPA-ready reports and CSV summaries

Architecture

Prototype architecture and data flow

Synthetic study data

250 sites → subjects → protocol visits

Protocol specification

Visit windows • dosing • medications • assessments • eligibility

Rules engine

Checks each visit and assigns Major / Minor / Administrative

Risk aggregation

Major rate + minor rate + 30-day trend + open CAPA burden

Web dashboard

Filters • sorting • site drawer • CAPA report • CSV export

Demo / Key Features

The strongest feature is the site drill-down from portfolio risk to individual deviations.

Portfolio view

Risk-tier filters, region search, sorting, risk scores and deviation counts.

Site drill-down

Shows subjects, visits, Major/Minor/Admin deviations and open CAPA items.

Actionable output

Each deviation includes a recommended corrective and preventive action.

Export

Site risk summary CSV for downstream analysis.

CAPA report

Generates a CAPA-ready text report for the selected site.

IBM Technologies

IBM technology mapping for the prototype

IBM Db2

Potential production data layer for structured study/site/visit data.

IBM Cloud

Potential deployment target for a scalable monitoring application and services.

Current prototype

Browser-based HTML/CSS/JavaScript implementation demonstrates the workflow and rule engine.

- Note: the uploaded prototype itself directly implements the monitoring logic in JavaScript; IBM services above are presented as production integration opportunities, not claimed as already connected.

Results / Impact

Success means faster detection, clearer prioritization and actionable follow-up.

250

Synthetic sites monitored

8

Regions represented

100

Maximum site risk score

3 levels

Major / Minor / Administrative

4 factors

Risk score components

2 exports

CSV summary + CAPA
report

- Impact: prioritize high-risk sites, investigate deviations by severity, and turn findings into corrective/preventive actions.
- The dataset is synthetic and the prototype rules/weights should be reviewed by clinical and regulatory leads before production use.

Team

Roles and contributions

HIYA SACHANIA

Role: Project lead / Problem
Research/Development

Built: problem framing, workflow
and presentation, Live dashboard

HIRVA KHANPARA

Role: Development

Built: rules engine, risk scoring
and dashboard logic

KESHA LADANI

Role: UI / UX

Built: Website layout, filters and
site drill-down

YASHVI BHIMANI

Role: Testing / documentation

Built: demo flow, outputs and
validation

Repository

Add your GitHub / repository
link here